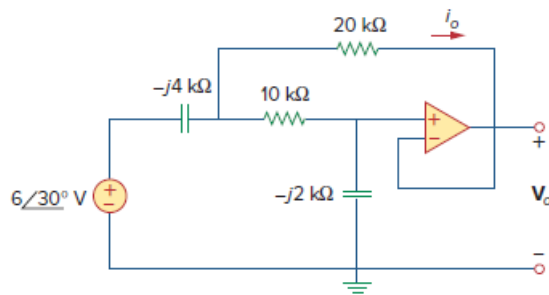


Problem

Determine V_o and I_o in the op amp circuit of Fig. 10.119.

Figure 10.119:



Step-by-step solution

Step 1 of 5

Refer to Figure 10.119 in the textbook.

Draw the circuit diagram with node voltages and currents.

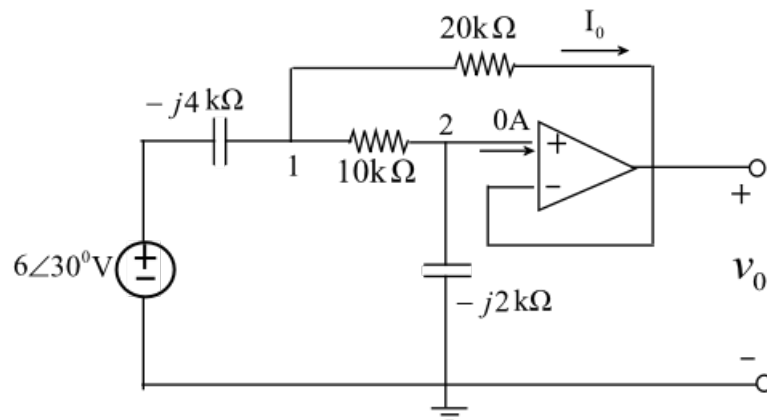


Figure 1

Step 2 of 5

Assume that the op-amp is ideal.

Currents through inverting and non-inverting terminals of an ideal op-amp are zero.

That is,

$$\begin{aligned}i_1 &= i_2 \\&= 0 \text{ A}\end{aligned}$$

The voltages at inverting and non-inverting nodes are equal.

That is,

$$v_2 = v_o$$

Apply Kirchhoff's current law at node 1.

$$\frac{V_1 - 6\angle 30^\circ}{-j4 \text{ k}} + \frac{V_1 - V_o}{10 \text{ k}} + \frac{V_1 - V_o}{20 \text{ k}} = 0$$

$$\frac{V_1 - 6\angle 30^\circ}{-j4} + \frac{V_1 - V_o}{10} + \frac{V_1 - V_o}{20} = 0$$

$$j5(V_1 - 6\angle 30^\circ) + 2(V_1 - V_o) + V_1 - V_o = 0$$

$$V_1(3 + j5) - 3V_o = 30\angle 120^\circ \dots\dots (1)$$

Step 3 of 5

Apply Kirchhoff's current law at node 2.

$$\frac{V_1 - V_o}{10} = \frac{V_o}{-j2} \quad [\text{Since, } V_2 = V_o]$$

$$(-j2)(V_1 - V_o) = 10V_o$$

$$(-j2)V_1 = V_o(10 - j2)$$

$$V_1 = V_o \left(\frac{10 - j2}{-j2} \right)$$

$$V_1 = V_o(1 + j5) \dots\dots (2)$$

Step 4 of 5

Substitute $V_o(1 + j5)$ for V_1 in equation (1).

$$V_o(1 + j5)(3 + j5) - 3V_o = 30\angle 120^\circ$$

$$V_o(-22 + j20) - 3V_o = 30\angle 120^\circ$$

$$V_o(-25 + j20) = 30\angle 120^\circ$$

$$V_o = \frac{30\angle 120^\circ}{-25 + j20}$$

$$= \frac{30\angle 120^\circ}{32\angle 141.3^\circ}$$

$$= 0.94\angle -21.3^\circ \text{ V}$$

Therefore, the output voltage, V_o is $\boxed{0.94\angle -21.3^\circ \text{ V}}$.

Step 5 of 5

From Figure 1,

$$I_0 = \frac{V_1 - V_0}{20 \text{ k}\Omega}$$

Substitute $V_0(1 + j5)$ for V_1 .

$$\begin{aligned} I_0 &= \frac{V_0(1 + j5) - V_0}{20 \text{ k}\Omega} \\ &= \frac{V_0(1 + j5) - V_0}{20 \text{ k}\Omega} \\ &= \frac{V_0(j5)}{20 \text{ k}\Omega} \end{aligned}$$

Substitute $0.94 \angle -21.3^\circ \text{ V}$ for V_0 .

$$\begin{aligned} I_0 &= \frac{(0.94 \angle -21.3^\circ)(5 \angle 90^\circ)}{20 \angle 0^\circ} \text{ mA} \\ &= 0.235 \angle 68.7^\circ \text{ mA} \end{aligned}$$

Therefore, the current, I_0 is $\boxed{0.235 \angle 68.7^\circ \text{ mA}}$.